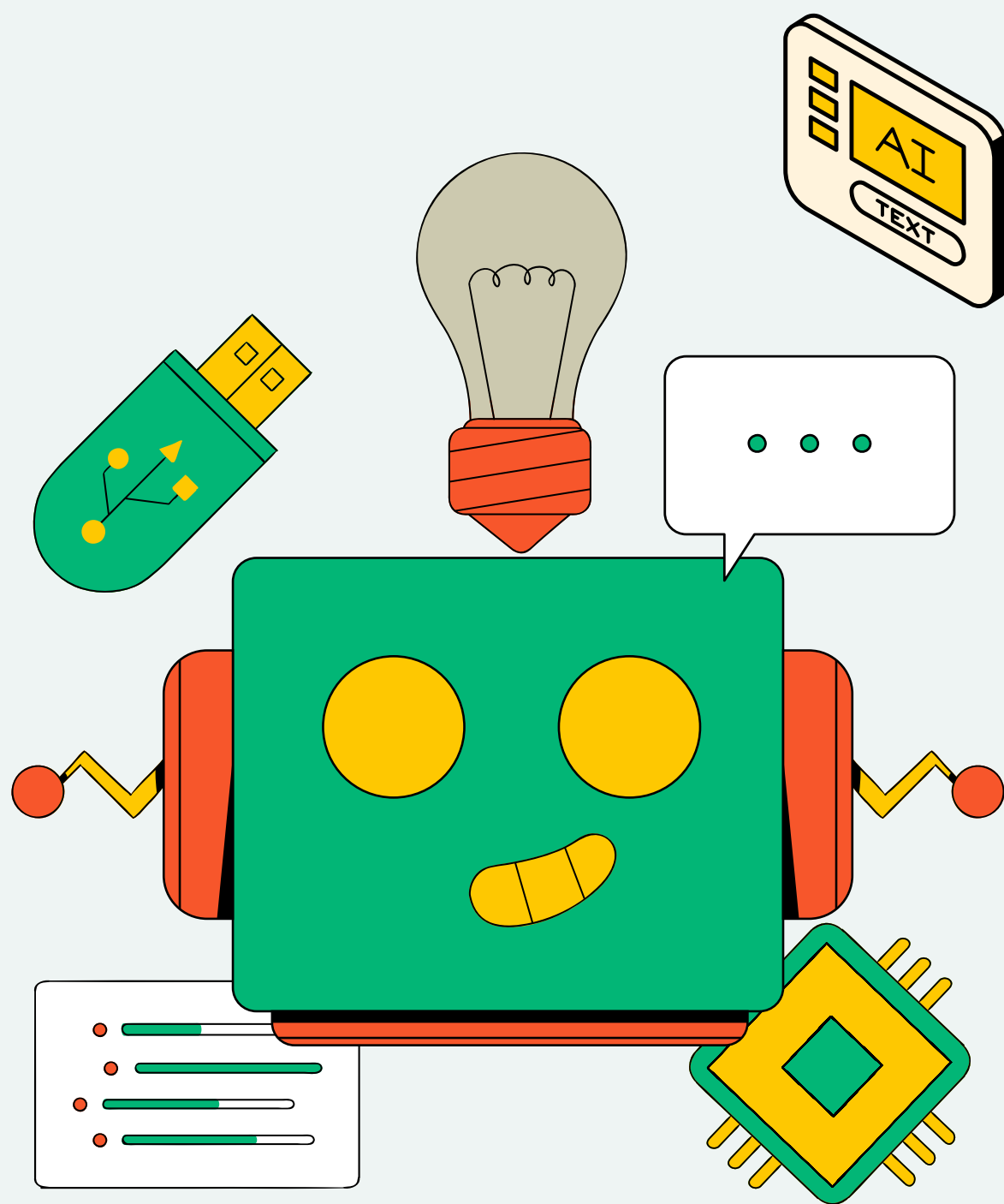


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A COMPARATIVE STUDY OF SUPERVISED AND SELF-SUPERVISED LEARNING FOR SEMG AND ACCELEROMETER-BASED HAND GESTURE RECOGNITION ON NINAPRO DB?

PRESENTED BY:

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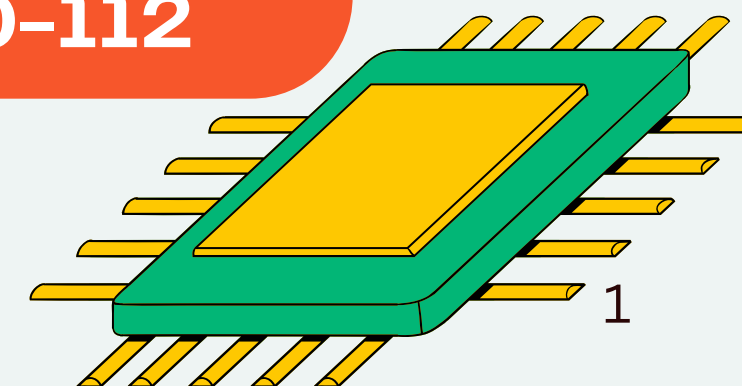
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ATKIYA MAISHA

2022-3-60-110

MOHUA AKTER

2022-3-60-112





PRESENTATION OUTLINE

- Introduction
- Dataset
- Related Work
- Supervised Learning Models
- Self-Supervised Learning Models
- The Best Supervised Method
- The Best Self-Supervised Method
- Learning Curves of the Best Supervised Method
- Pretraining loss Curve of the Best self-Supervised Method
- T-SNE Graphs of All Self-Supervised Methods
- Conclusion
- References



INTRODUCTION

This project focuses on hand gesture recognition using the NinaPro DB7 dataset with sEMG and accelerometer signals. Supervised and self-supervised learning methods are explored under different train-test splits and evaluated using standard classification metrics.



DATASET

Dataset: NinaPro DB7 <https://ninapro.hevs.ch/instructions/DB7.html>

Subjects: 22 total (20 intact, 2 amputees)

Sensors: 12 Delsys Trigno sensors with sEMG + inertial signals

Used in this work: 12-channel EMG + 12×3 accelerometer signals

Gesture subset: Gesture B, classes 13–29

Protocol: 5 s movement + 3 s rest, 6 repetitions

Total windows: 87,614



RELATED WORK

Chen et al. (2025) proposed MFF-Net, which combines CNN, BiLSTM, and attention to extract time, frequency, and spatial features, and achieved 92.47% accuracy on NinaPro DB7.

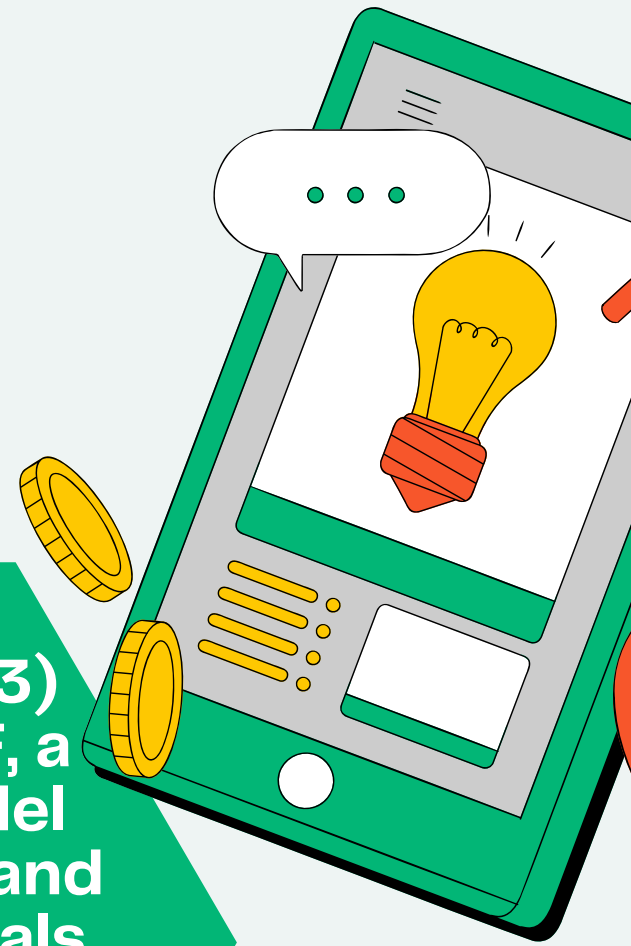
Duan et al. (2023) proposed HyFusion, a multimodal fusion framework using sEMG + ACC, and reported 96.44% accuracy on NinaPro DB7.

Fatayer et al. (2022) introduced ALR-CNN, which refines noisy labels during training, and achieved 95.85% accuracy on NinaPro DB7.

Sri-iesaranusorn et al. (2021) used a deep neural network/CNN baseline for 41-class gesture recognition and obtained 91.69% accuracy on NinaPro DB7.

Fang et al. (2023) proposed MMDF, a deep forest model that fuses sEMG and acceleration signals, and achieved 91.40% accuracy on healthy NinaPro DB7 subjects.

Fasulo et al. (2026) proposed TinyMyo, a lightweight transformer-based foundation model, which achieved 89.4% on NinaPro DB5 and 96.7% on EPN-612.



SUPERVISED LEARNING METHODS



TinyMyo

Deep Neural Network

DeepCNN

**Multi-Kernal Attention CNN/ MKA-
CNN (Custom CNN)**



SUPERVISED LEARNING METHODS

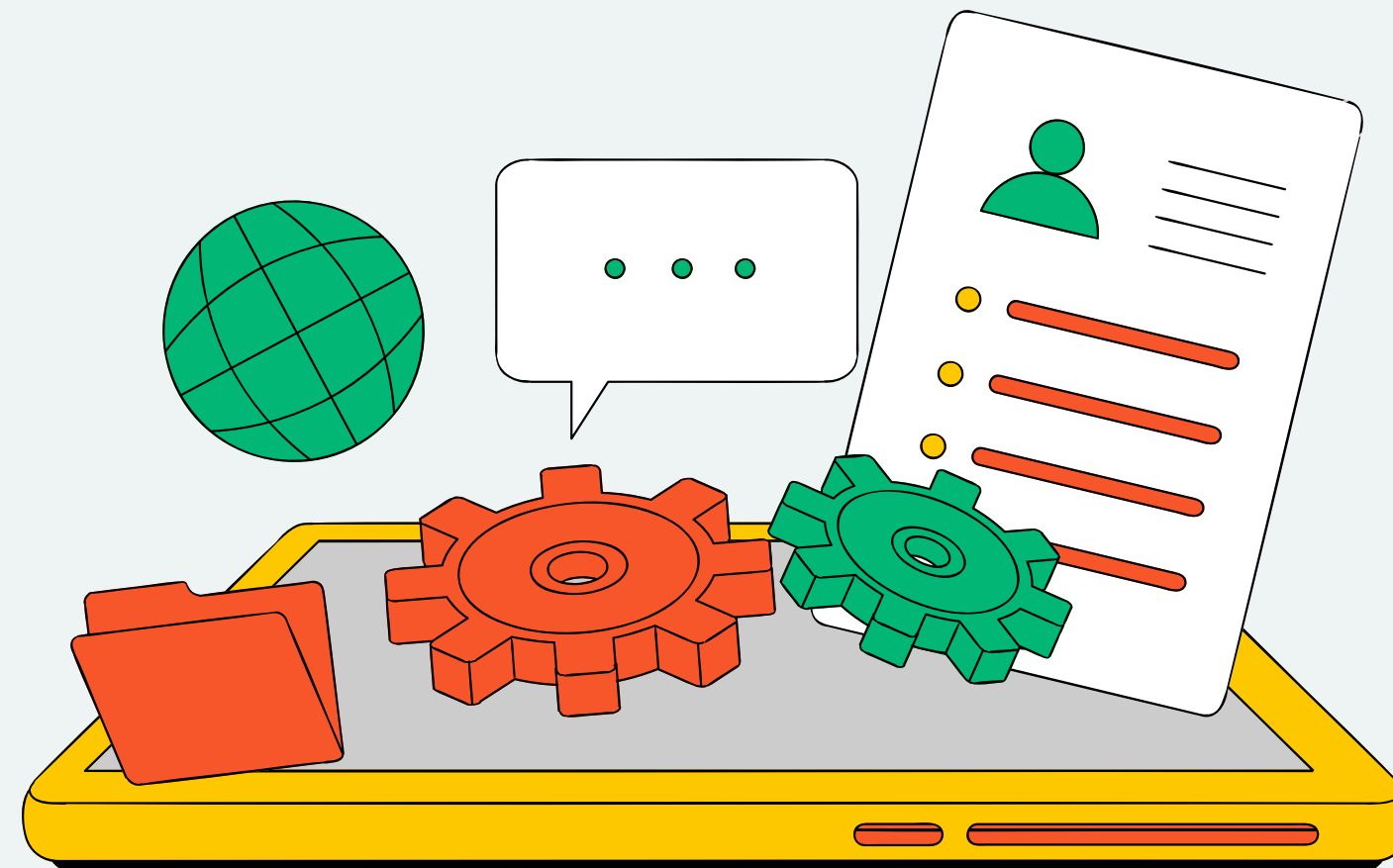
Model	Best Split	Accuracy	Precision	Recall	F1-Score
TinyMyo	90:10	97.81%	97.83%	97.81%	97.81%
DNN	90:10	92.83%	92.87%	92.83%	92.83%
DeepCNN	90:10	98.03%	98.04%	98.03%	98.03%
MKA 1D-CNN	80:20	98.76%	98.76%	98.76%	98.76%

SELF-SUPERVISED LEARNING METHODS

BYOL – Bootstrap Your Own Latent

TNC – Temporal Neighborhood Coding

SimCLR – A Simple Framework for Contrastive Learning of Visual Representations.

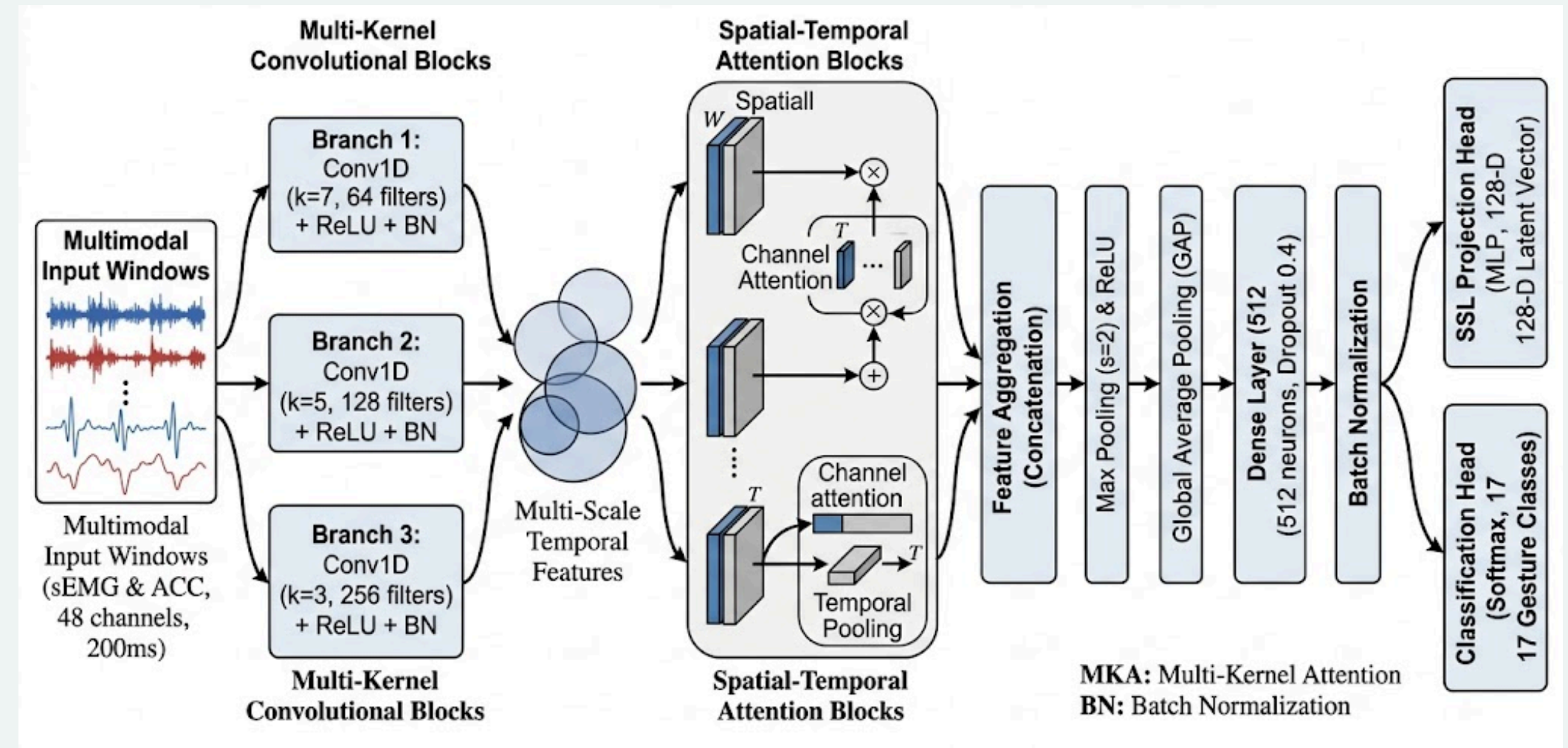


SELF-SUPERVISED LEARNING METHODS

SSL Method	Best Encoder	Accuracy	Precision	Recall	F1-Score
SimCLRv1	ResNet1D	98.33%	98.34%	98.33%	98.33%
SimCLRv2	Custom CNN 1D	98.36%	98.36%	98.36%	98.36%
BYOL	Custom CNN 1D	98.92%	98.92%	98.92%	98.92%
TNC	Custom CNN 1D	98.68%	98.68%	98.68%	98.68%

THE BEST SUPERVISED METHOD

- **MULTIMODAL INPUT:** USES BOTH SEMG AND ACCELEROMETER SIGNALS
- **MULTI-KERNEL CONVOLUTION:** PARALLEL CONV1D BRANCHES CAPTURE MULTI-SCALE TEMPORAL FEATURES
- **ATTENTION MECHANISM:** SPATIAL-TEMPORAL ATTENTION HIGHLIGHTS IMPORTANT CHANNELS AND TIME REGIONS
- **FEATURE AGGREGATION:** COMBINES REFINED FEATURES INTO A COMPACT REPRESENTATION
- **CLASSIFICATION HEAD:** DENSE LAYERS CLASSIFY THE INPUT INTO 17 GESTURE CLASSES
- **SSL PROJECTION HEAD:** LEARNS A 128-DIMENSIONAL LATENT REPRESENTATION FOR SELF-SUPERVISED LEARNING



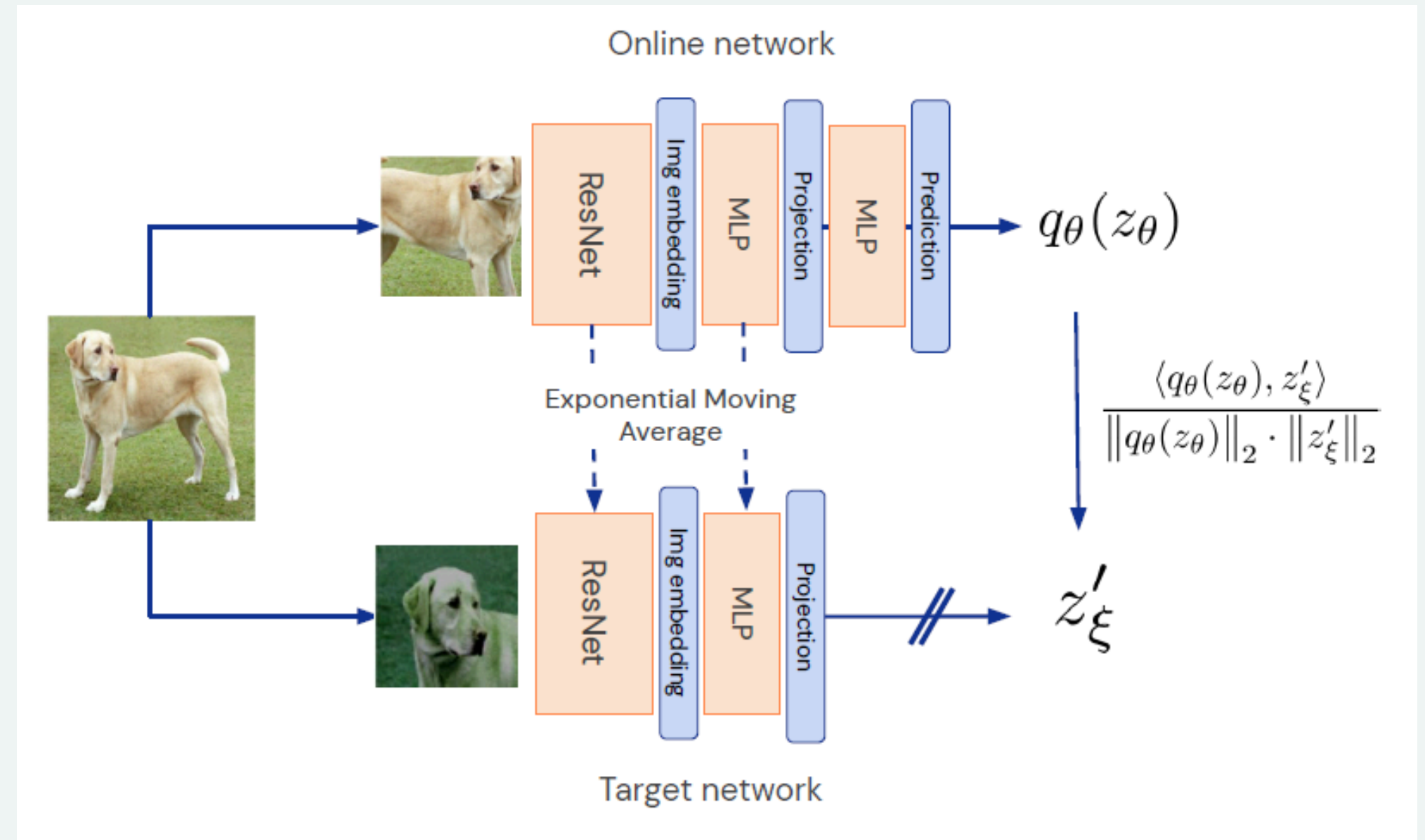
Architecture of MKA-CNN

PERFORMANCE OF THE BEST SUPERVISED METHOD ACROSS TRAIN-TEST SPLITS

Split	Accuracy	Precision	Recall	F1-Score
90:10	98.72%	98.73%	98.72%	98.72%
80:20	98.76%	98.76%	98.76%	98.76%
70:30	97.93%	97.94%	97.93%	97.93%
60:40	97.67%	97.68%	97.67%	97.67%
50:50	95.50%	95.54%	95.50%	95.50%
40:60	96.15%	96.16%	96.15%	96.15%
30:70	93.90%	93.93%	93.90%	93.90%
20:80	93.08%	93.08%	93.08%	93.07%
10:90	88.75%	88.82%	88.75%	88.76%

BEST SELF-SUPERVISED METHOD

- BYOL (BOOTSTRAP YOUR OWN LATENT) WORKS BY CREATING TWO AUGMENTED VIEWS OF THE SAME INPUT AND PASSING THEM THROUGH AN ONLINE NETWORK AND A TARGET NETWORK.
- THE ONLINE NETWORK TRIES TO PREDICT THE TARGET REPRESENTATION, WHILE THE TARGET NETWORK IS UPDATED USING AN EXPONENTIAL MOVING AVERAGE. THIS ALLOWS THE MODEL TO LEARN STRONG EMBEDDINGS WITHOUT USING NEGATIVE SAMPLES, MAKING BYOL EFFECTIVE FOR REPRESENTATION LEARNING AND DOWNSTREAM CLASSIFICATION TASKS.



PERFORMANCE OF THE BEST SELF-SUPERVISED METHOD ACROSS TRAIN-TEST SPLITS

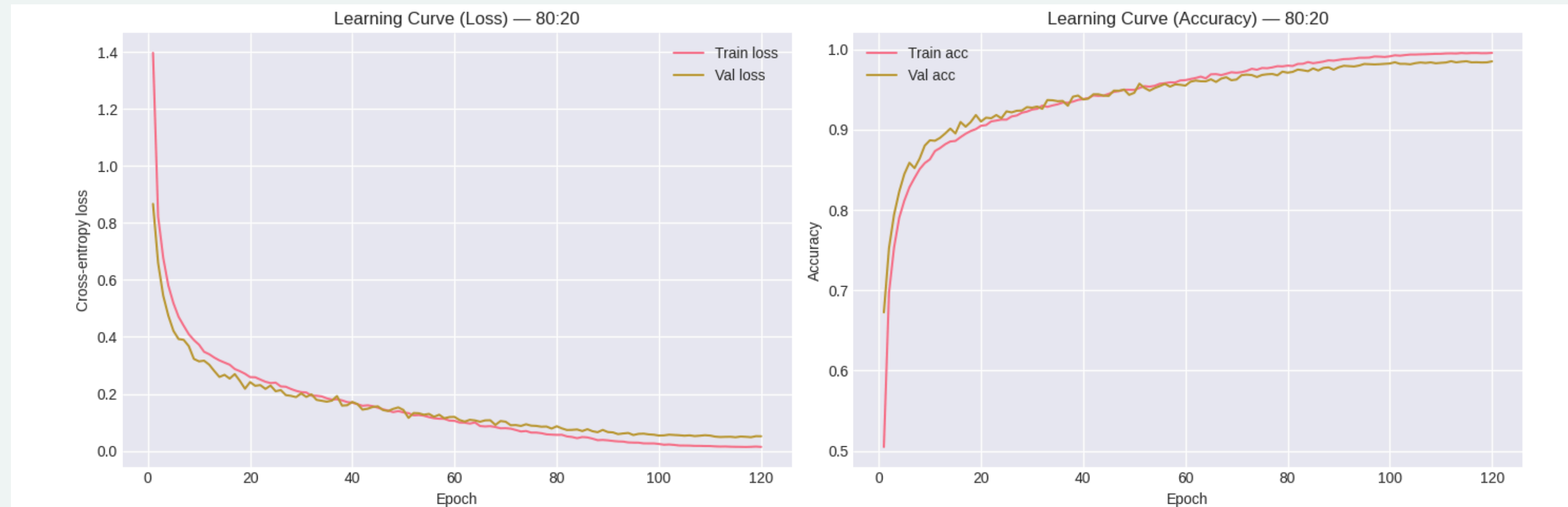
Train:Test Split	Accuracy	Precision	Recall	F1-Score
90:10	98.92%	98.92%	98.92%	98.92%
80:20	98.54%	98.54%	98.54%	98.54%
70:30	97.99%	97.99%	97.99%	97.99%
60:40	97.21%	97.22%	97.21%	97.21%
50:50	96.78%	96.79%	96.79%	96.79%
40:60	95.68%	95.69%	95.68%	95.68%
30:70	93.66%	93.66%	93.66%	93.65%
20:80	91.88%	91.89%	91.88%	91.88%
10:90	86.01%	86.06%	86.01%	85.99%

BYOL ACCURACY ABLATION: EPOCHS VS. SPLIT RATIO

THE RESULTS INDICATE THAT BYOL PERFORMS BEST AT THE 90:10 SPLIT, REACHING 98.92% ACCURACY WITH 250 PRE-TRAINING EPOCHS. ACCURACY DECREASES AS TRAINING DATA IS REDUCED, BUT BYOL REMAINS STRONG ACROSS ALL SPLITS, SHOWING ROBUST AND STABLE REPRESENTATION LEARNING.

Train:Test Split	250 Epochs	200 Epochs	150 Epochs
90:10	98.92%	98.77%	98.72%
80:20	98.54%	98.56%	98.45%
70:30	97.99%	97.81%	97.85%
60:40	97.21%	97.08%	96.93%
50:50	96.78%	96.70%	96.71%
40:60	95.68%	95.58%	95.49%
30:70	93.66%	93.55%	93.51%
20:80	91.88%	91.75%	92.16%
10:90	86.01%	86.90%	86.56%

LEARNING CURVES OF THE BEST SUPERVISED METHOD



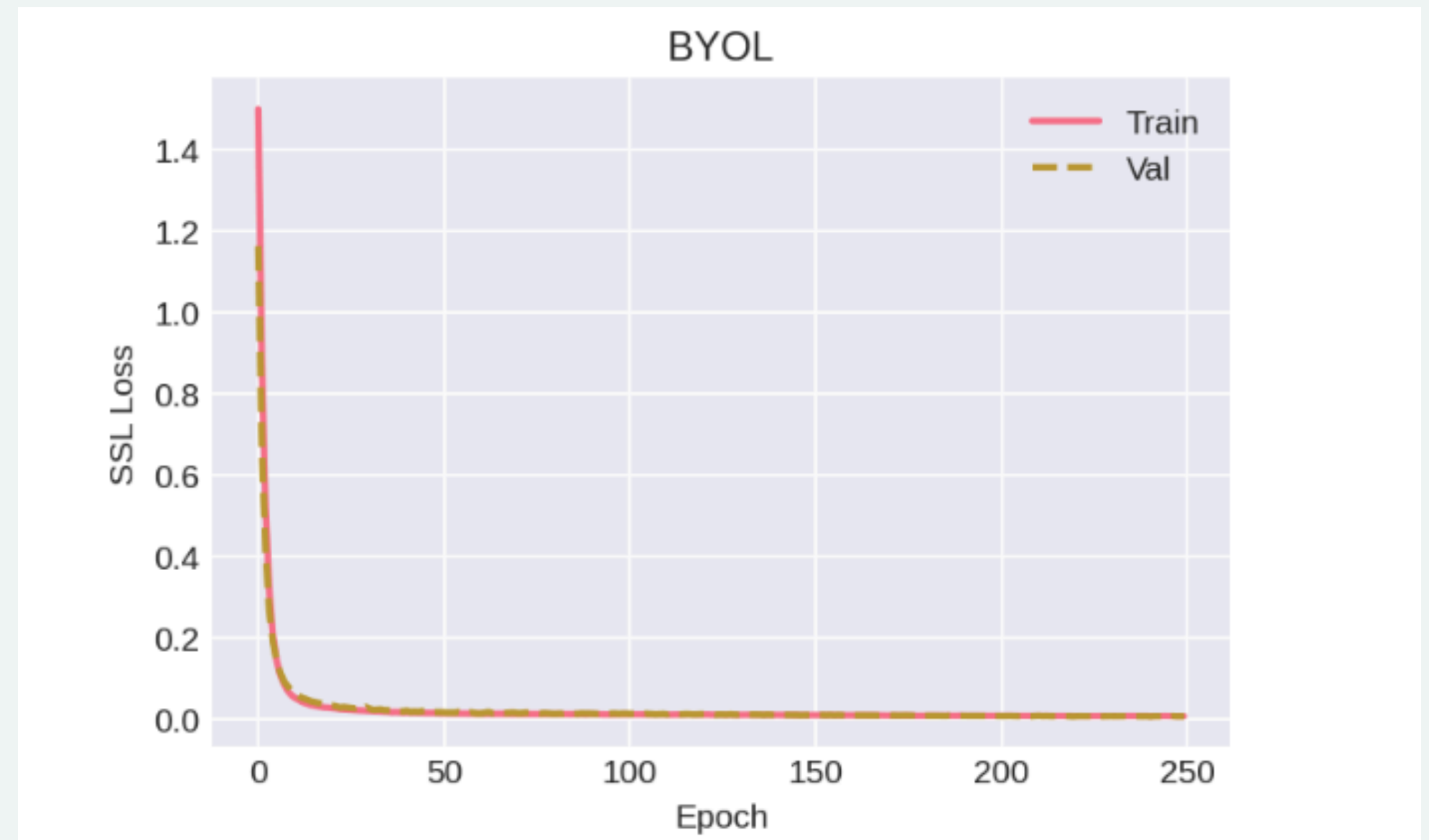
THE LEARNING CURVES SHOW STABLE TRAINING ON THE 80:20 SPLIT, WITH DECREASING LOSS AND INCREASING ACCURACY FOR BOTH TRAINING AND VALIDATION, INDICATING GOOD CONVERGENCE AND MINIMAL OVERFITTING.

PRETRAINING LOSS CURVE OF THE BEST SELF-SUPERVISED METHOD

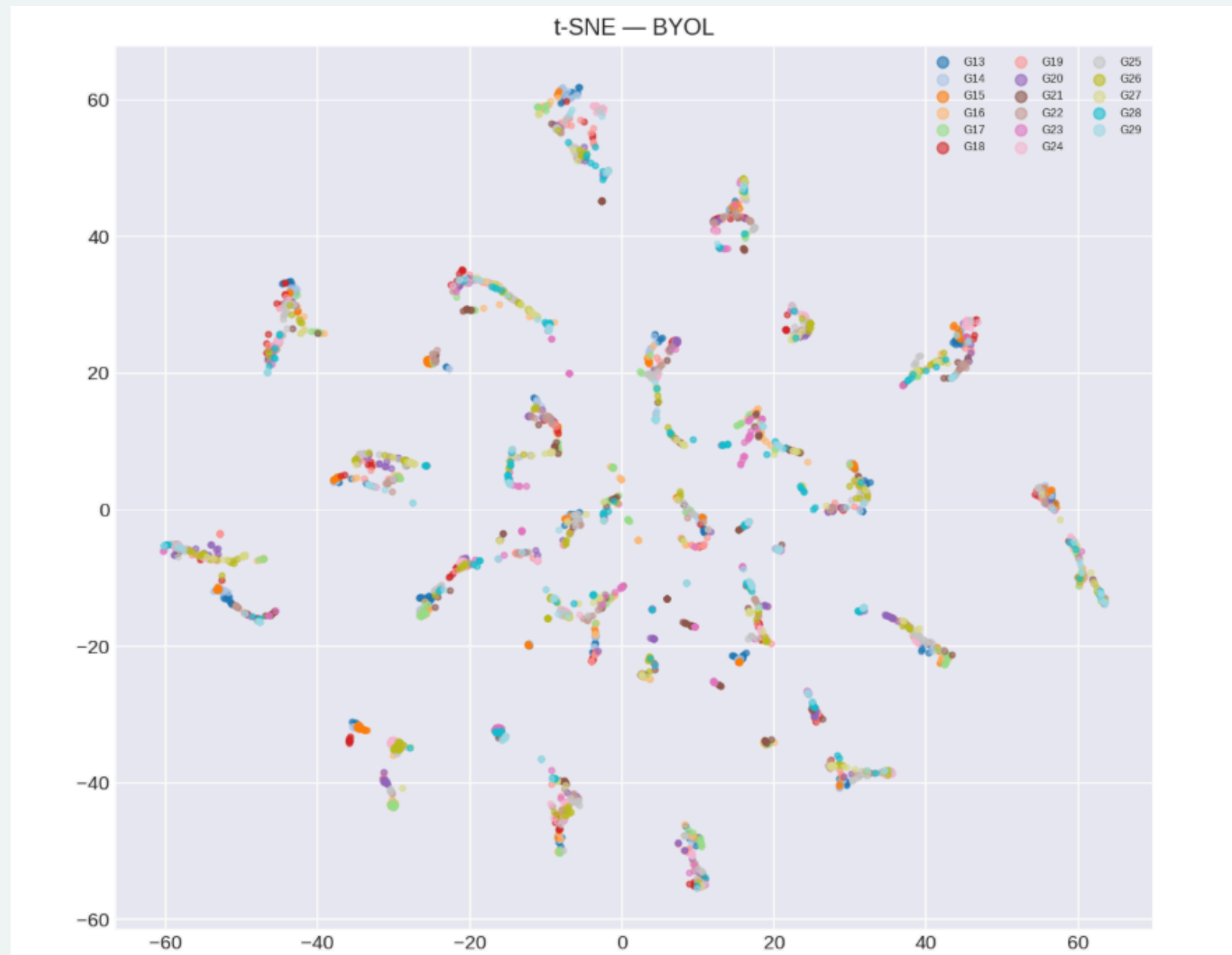
THE CURVE SHOWS THAT BYOL LEARNS VERY QUICKLY IN THE EARLY EPOCHS, SINCE BOTH TRAINING AND VALIDATION SSL LOSS DROP SHARPLY AT THE BEGINNING AND THEN GRADUALLY FLATTEN NEAR ZERO.

THIS MEANS:

- THE MODEL IS CONVERGING WELL
- TRAIN AND VALIDATION CURVES ARE VERY CLOSE, SO LEARNING IS STABLE
- THERE IS NO CLEAR SIGN OF OVERFITTING DURING PRETRAINING



T-SNE GRAPHS OF ALL SELF-SUPERVISED METHODS



THIS T-SNE PLOT OF BYOL EMBEDDINGS SHOWS HOW THE LEARNED FEATURE REPRESENTATIONS ARE ARRANGED IN 2D.

WHAT IT INDICATES:

- THE POINTS FORM MANY SEPARATE CLUSTERS, SO BYOL HAS LEARNED STRUCTURED FEATURE REPRESENTATIONS.
- HOWEVER, THE COLORS FOR DIFFERENT GESTURE CLASSES ARE STILL MIXED WITHIN MANY CLUSTERS, MEANING THE CLASSES ARE NOT PERFECTLY SEPARATED IN THE RAW EMBEDDING SPACE.
- THIS SUGGESTS BYOL LEARNED USEFUL GENERAL FEATURES, BUT THE EMBEDDINGS STILL NEED A CLASSIFIER OR FINE-TUNING FOR BETTER CLASS DISCRIMINATION.

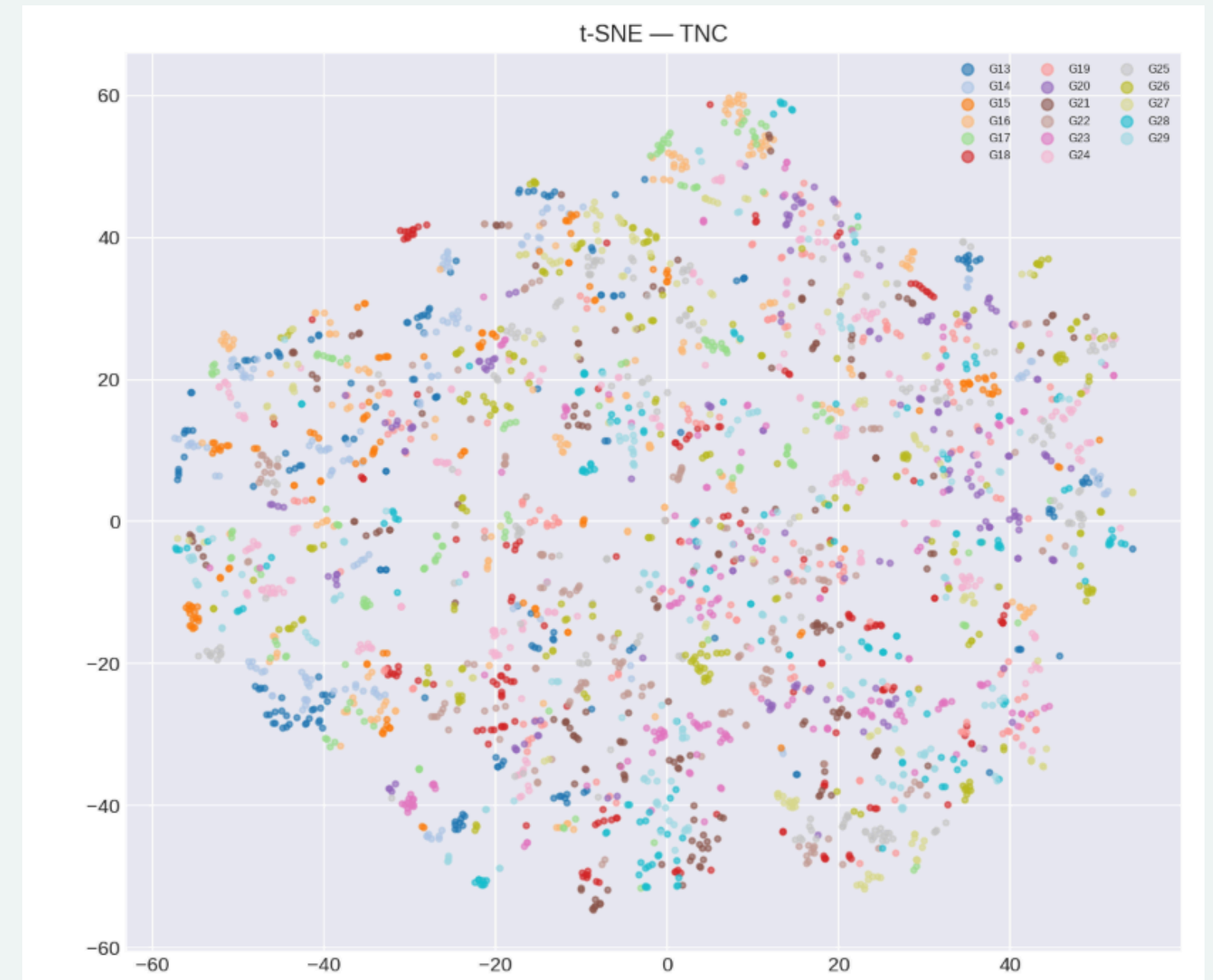
BYOL

T-SNE GRAPHS OF ALL SELF-SUPERVISED METHODS

THIS T-SNE PLOT FOR TNC EMBEDDINGS SHOWS THAT THE LEARNED FEATURES ARE MORE SPREAD OUT BUT STILL HIGHLY MIXED ACROSS CLASSES.

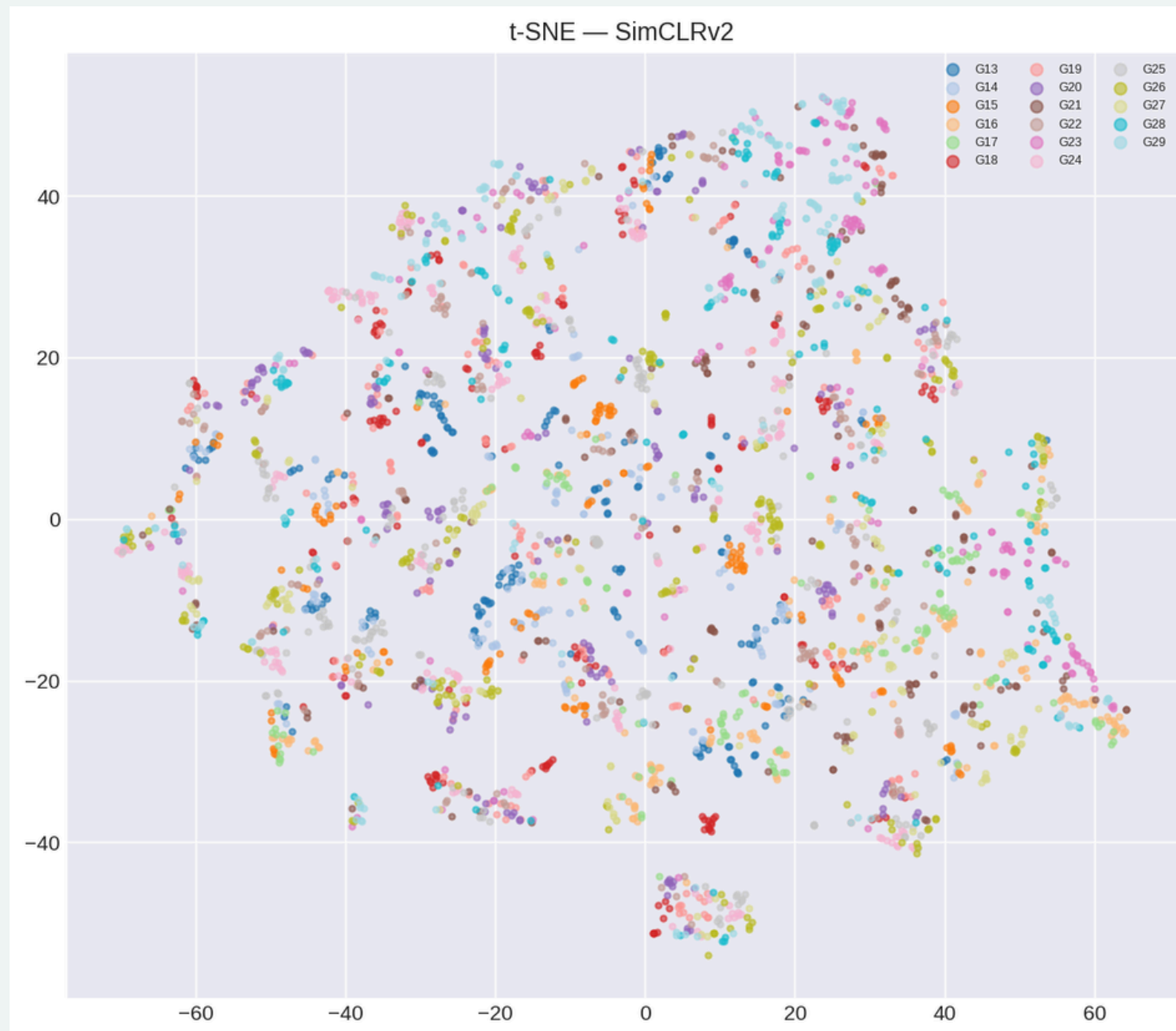
WHAT IT SUGGESTS:

- EACH POINT IS A SAMPLE EMBEDDING LEARNED BY TNC.
- THE PLOT FORMS A BROAD GLOBAL STRUCTURE, SO TNC IS CAPTURING MEANINGFUL PATTERNS IN THE DATA.
- BUT MANY COLORS ARE OVERLAPPING HEAVILY, WHICH MEANS DIFFERENT GESTURE CLASSES ARE NOT CLEARLY SEPARATED IN THE RAW FEATURE SPACE.
- COMPARED WITH A WELL-SEPARATED EMBEDDING PLOT, THIS INDICATES THE LEARNED REPRESENTATIONS ARE USEFUL BUT NOT STRONGLY CLASS-DISCRIMINATIVE BY THEMSELVES.



TNC

T-SNE GRAPHS OF ALL SELF-SUPERVISED METHODS



THIS T-SNE PLOT FOR SIMCLRv2 EMBEDDINGS SHOWS THAT THE MODEL LEARNS STRUCTURED FEATURE GROUPS, BUT THE GESTURE CLASSES ARE STILL PARTIALLY OVERLAPPING.

WHAT IT SUGGESTS:

- THE POINTS FORM MANY LOCAL CLUSTERS, SO SIMCLRv2 IS CAPTURING USEFUL PATTERNS FROM THE SIGNALS.
- SOME SAME-COLOR POINTS STAY RELATIVELY CLOSE, WHICH MEANS SAMPLES FROM THE SAME GESTURE CAN SHARE SIMILAR EMBEDDINGS.
- BUT THERE IS STILL NOTICEABLE MIXING BETWEEN COLORS, SO THE CLASSES ARE NOT FULLY SEPARATED IN THE RAW EMBEDDING SPACE.
- THIS MEANS SIMCLRv2 LEARNS MEANINGFUL REPRESENTATIONS, BUT FINE-TUNING OR A STRONGER CLASSIFIER IS STILL NEEDED FOR BETTER GESTURE DISCRIMINATION.

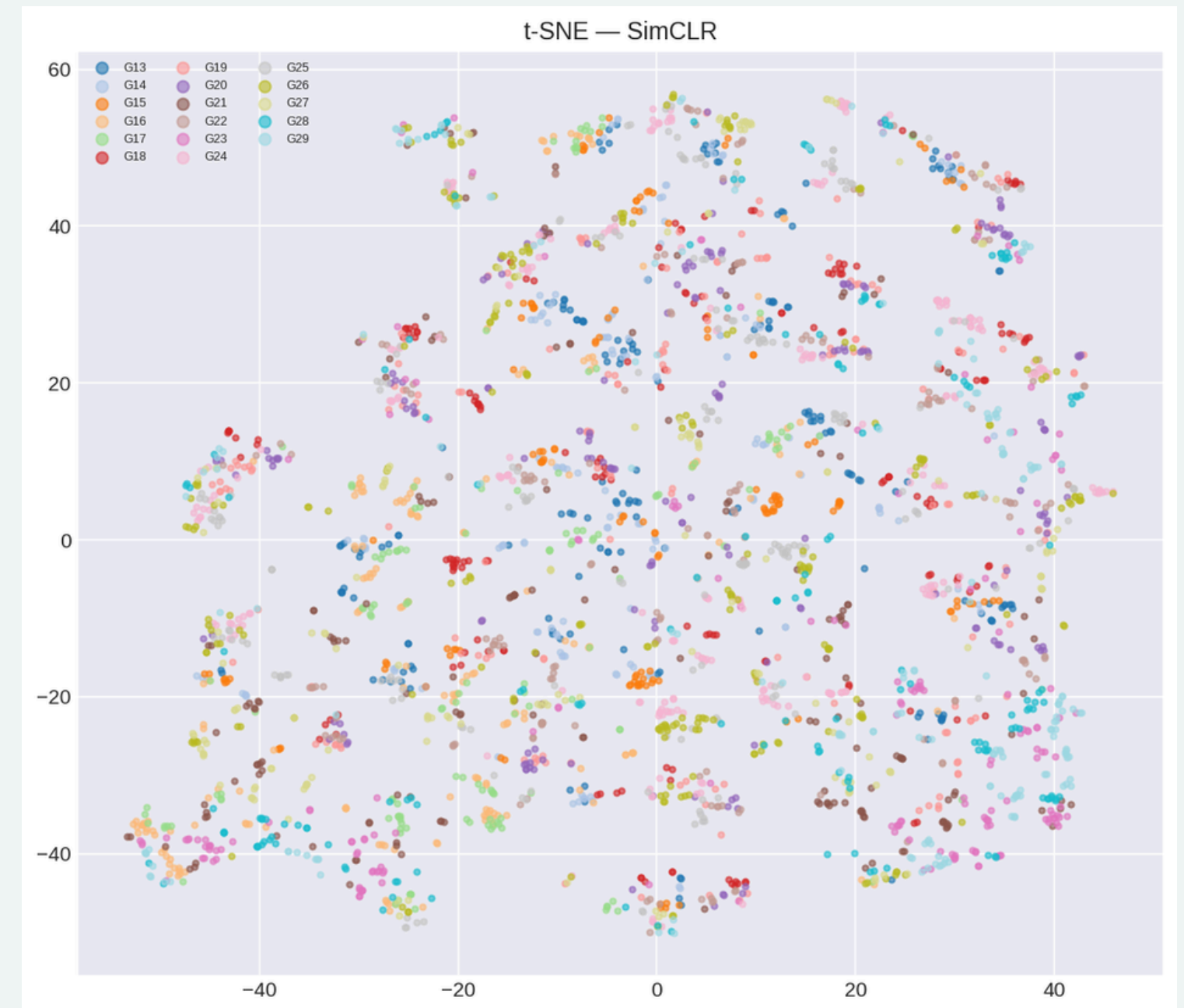
SIMCLRv2

T-SNE GRAPHS OF ALL SELF-SUPERVISED METHODS

THIS T-SNE PLOT FOR SIMCLR EMBEDDINGS SHOWS THAT THE MODEL LEARNS SOME STRUCTURE IN THE FEATURE SPACE, BUT THE GESTURE CLASSES STILL OVERLAP A LOT.

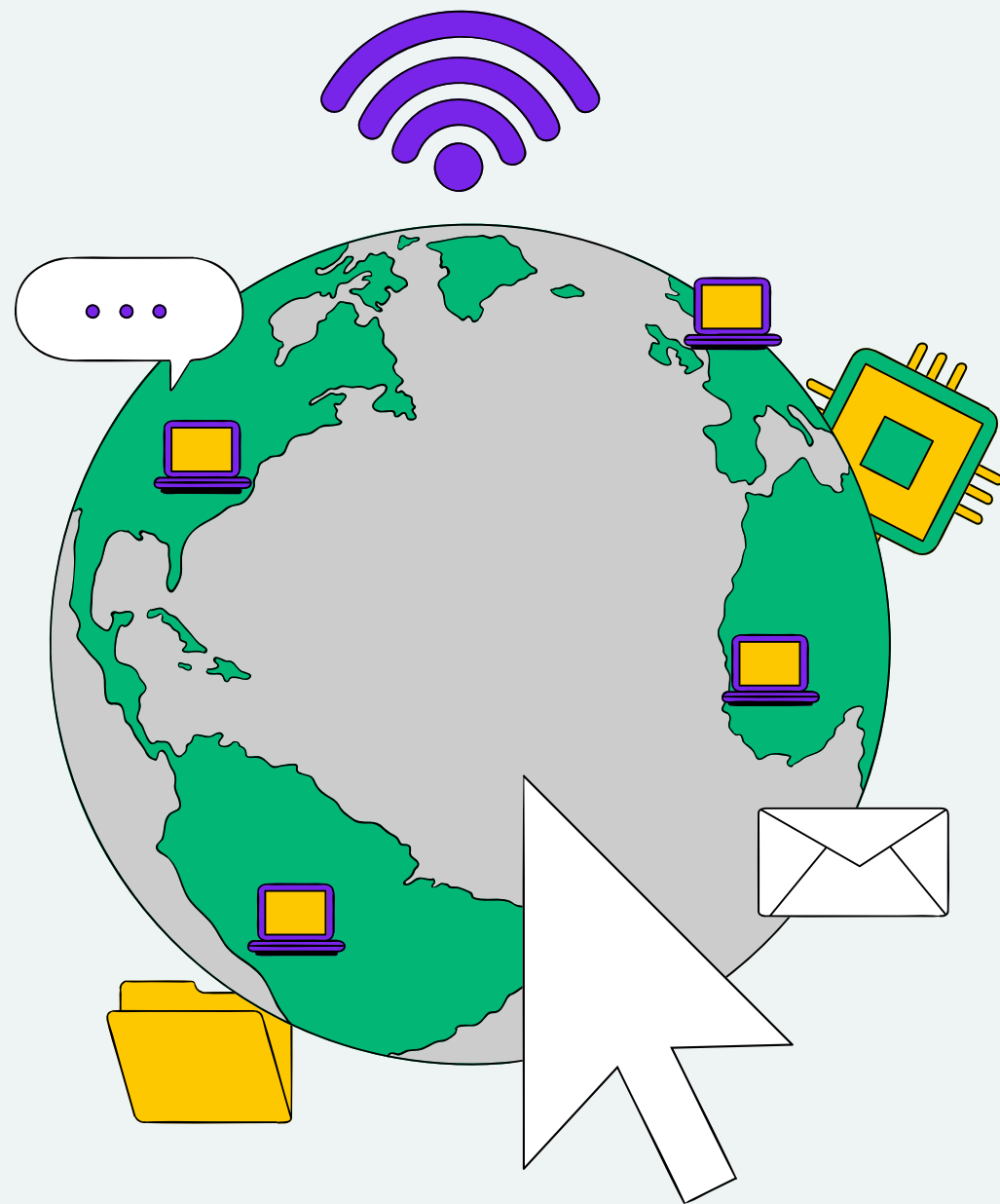
WHAT IT SUGGESTS:

- THE POINTS FORM SEVERAL LOCAL GROUPS/CLUSTERS, SO SIMCLR IS LEARNING USEFUL PATTERNS.
- SOME SAMPLES FROM THE SAME GESTURE STAY CLOSE TOGETHER.
- BUT MANY COLORS ARE MIXED ACROSS CLUSTERS, MEANING THE LEARNED EMBEDDINGS ARE NOT CLEARLY CLASS-SEPARATED.
- SO, SIMCLR LEARNS MEANINGFUL FEATURES, BUT THE RAW REPRESENTATION IS NOT STRONGLY DISCRIMINATIVE ON ITS OWN.



SimCLRv1

CONCLUSION



This project compared supervised and self-supervised methods for hand gesture recognition on NinaPro DB7 using sEMG and accelerometer signals. MKA-CNN achieved the best supervised performance, while BYOL was the strongest self-supervised method. The study shows that self-supervised learning can learn effective representations and is a promising direction for robust gesture recognition.



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